

Context utility depends on its readout

Central finding: during matched training, native inference reverses its preference for support context while a fixed prototype does not. The end-to-end effect of graph context therefore cannot be read as the usefulness of the information it encodes.

A A real but non-universal phenomenon

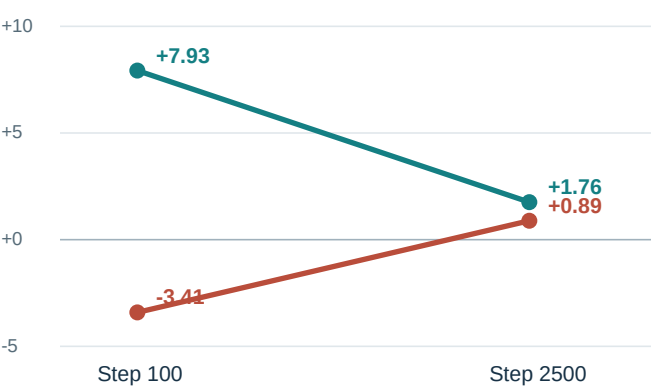
Support suppression: pooled AUC change, points

	Pol.	Elect.	Pages	Bots	Susp.
COVID	-0.1*	+0.0*	+0.1	-4.0	+1.2
Political	-5.7	-0.4	-3.9	-2.4	-2.8
Hong Kong	+9.9	+2.0	+2.5	-7.5	+0.6*
Election	-0.2	-0.0*	-1.3	-10.1	+0.2*
Pages	+1.8	+1.3	+0.9	-1.6	+1.4
Midterm	+3.1	+0.3	+0.8	-0.6	-0.1*
Bots	+0.8	+0.1	+0.1*	-2.6	+0.2*
Ukraine	-0.1	-0.0	-0.0*	-5.4	+1.7
Suspended	+7.4	+0.1*	-0.6	-8.3	+1.4

Nine sources, five targets; mean of two episode streams, not training seeds. Teal helps, red hurts; * signs differ. Hong Kong is an extreme, not the only affected source.

B Training reverses only the native preference

Page suppression gain: within-episode AUC points



Same page episodes, 3 seeds x 2 streams; mean effects. Prototypes improve with suppression in all six comparisons at both steps. Native: 5/6 harmed early, 6/6 helped later. Both improve absolutely. **The suppressed path catches up; intact inference does not deteriorate.**

C Similar AUC gains need not mean repair

Political suppression gain; within-episode AUC points

Checkpoint	Current init.	Train table
2.5k original	+7.25	+7.26
50k original	+8.76	+0.34
50k fresh	+10.35	+1.87

Early suppression improves accuracy under either initialization; late suppression hurts it under both. Restoring the training label table strongly moderates the late AUC gain. Early lower sensitivity was predicted before the new factorial outcomes.

Late suppressed supports collapse upstream of the readout. Removing the learned default message does not restore downstream geometry or improve intact inference. **No offset-based repair is supported.**

D Generality result: both roles can help

Public Wiki to FB15K-237: accuracy (%)

Intact	73.8
Raw prototype	72.4
Untrained	69.8
Suppress supports	39.9
Suppress queries	39.0

A useful original-style checkpoint needs both roles: support/query suppression costs 33.87/34.80 points. 500 paired 20-way episodes, 8,001 updates. This contradicts universal support harm. Public queries are not fixed under the role intervention; do not claim the social mediation result generalizes.

Contribution: matched graph-role interventions show why native context utility must be distinguished from information available to a fixed readout. At page step 100, suppressed prototype AUC is .649 versus native .504; the information exists but native inference does not exploit it. This extends beyond a static graph-distance account, without claiming a universal predictor or isolating one defective projection.

Decision: keep the explanatory paper direction. The encoder-solver isolation study already running tests a training consequence; no outcome is yet established. Close inference-control expansion. Query processing also differs between the readouts, so the precise learned source of their mismatch remains unresolved.

Evidence: matched page prototype (8ccd3686); nine-source map; initialization (9e19e5eb); offset (48907a3b); public quality/roles. Public logit-tolerance replay failed; an explicit amendment before scientific outcomes required exact labels and all 40,000 predicted classes, which matched. No tolerances were enlarged. These are conditional results, not seed-level generality.